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ORIGINAL RESEARCH

Predicting the Academic Performance of the Students: Machine Learning Algorithms

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Abstract
Schools and colleges use students' grades as a primary way to evaluate how well their teachers are performing and to assess the overall quality of their student body. Academic and practical performance are two key indicators of a student's success in school. Most research suggests that students share common traits while overlooking their unique characteristics. The main aim of this paper is to develop a framework for understanding how generative artificial intelligence (AI) technologies can enhance classroom learning and support students' success. On the other hand, understanding what each student learns best depends on their individual characteristics, which are important for improving academic performance. This study employs machine learning (ML) techniques, including Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), K-nearest neighbors (KNN), and Support Vector Machine (SVM) to classify students and predict their performance. Neural networks (NN) are used to achieve this goal. We tested the algorithms on a standard dataset of 2,392 instances of high school students from Kaggle. We analyzed students' scores to see how they fit into different groups. The method involves handling missing values, normalizing the data, and adjusting variables as necessary. The RF selection method identified the most important traits. The GB method achieved 92% accuracy on the test, while the NN predicted student performance with an MSE of 0.0665 and an R^2 of 93.27%.

Keywords Machine learning · Artificial intelligence · Students · Prediction · Classification

Introduction

To improve education, we need accurate methods to predict and analyze student performance in school. With big data technology and the spread of educational informatization, we can now record increasingly multidimensional spatio-temporal data. This presents new challenges for teachers but also provides them with a wealth of information [1]. Outcomes may not effectively support both students and instructors, as traditional predictive methods fail to capture the many factors influencing students' academic success [2]. It is crucial to use quantitative analysis and techniques to assess academic performance so that struggling students can be quickly identified and helped, thereby improving education quality and learning outcomes.

Educational Data Mining (EDM) has developed from applying data mining techniques within the educational field, aiming to clarify the complex links between students' academic results and the factors that affect them. EDM is a specialized form of data mining that seeks to extract useful information from educational data to help understand how teaching and learning function and to improve decision-making in education. Universities around the world are working to tackle challenges such as low student performance and high dropout rates [3–5]. The EDM field employs methods from various disciplines to achieve its objectives. The data mining cycle, as shown in Fig. 1, follows a structured process to extract meaningful insights from unstructured data.

Teachers often emphasize that forecasting students' academic success is crucial for taking preventive measures to boost learning and decrease the likelihood of failure and dropping out. It is necessary to analyze many elements to predict SAP. Currently, researchers typically use student

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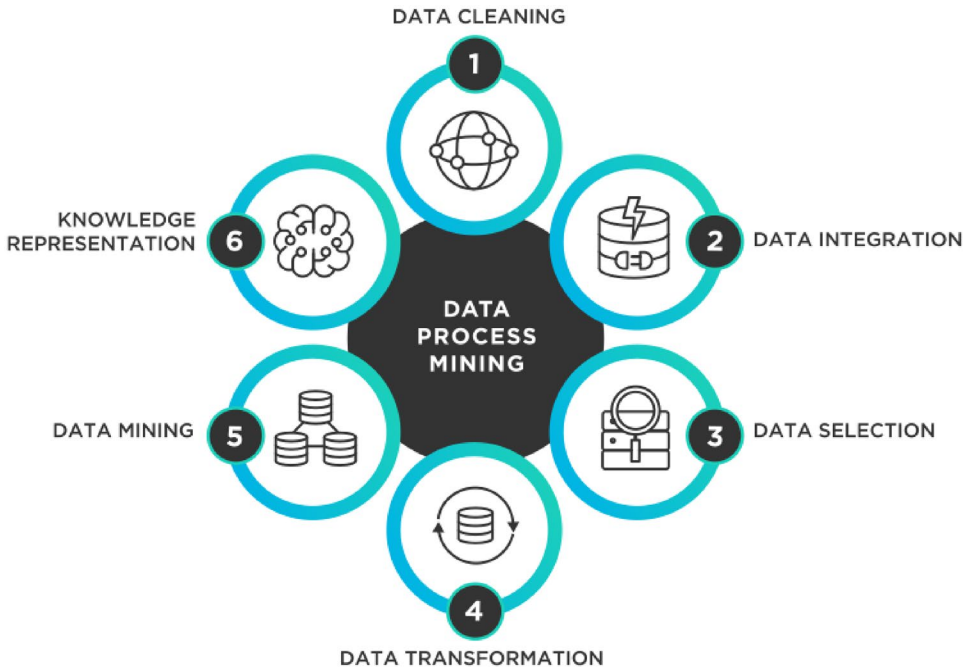


Fig. 1 DM cycle for processing the dataset [4]

records stored in administrative systems to forecast academic performance [6]. These records include basic student information and educational history. For instance, between 2016 and 2019, a public institution in Saudi Arabia had 2039 students registered in the Computer Science and Information Department. Mengash et al. [7] utilized data mining techniques to construct this information. Assisting with admissions decisions and making academic achievement predictions were the two main objectives. Using the MapReduce architecture, Baruah et al. [8] presented a method for forecasting students' performance. Feng et al. [9] applied discriminant analysis to assess clustering within the K-Means algorithm for evaluating and forecasting students' academic performance and to identify the optimal number of clusters. Utilizing data from educational technologies and online learning platforms, Liu and colleagues [10] constructed a feedforward spiking neural network to forecast student performance. Yue et al. [11] suggest using an ML model for predicting student performance.

The primary goal of the study is to identify common traits among different student groups to predict their success. However, this study often overlooks the unique characteristics each learner has. It highlights three main concerns. Most studies evaluate students' academic performance solely based on their overall grades [12]. Using a single metric to assess student learning introduces significant challenges due to the complexity of the concept. This approach makes it challenging to accurately measure student learning [13–15].

Problem Statement

This paper seeks to analyze the difficulties related to the quality of educational delivery systems in schools, colleges, and other educational institutions. This research examines the monitoring of academic performance to mitigate student dropout rates. This research will enable subsequent large-scale studies utilizing diverse algorithms on more extensive datasets, thereby illustrating the feasibility of broad application within the education sector.

Contribution

The paper provides a comprehensive overview of its key contributions as follows.

- We address the challenge of using ML techniques to predict the performance of specific nodes to enhance their effectiveness.
- The student's work was graded on a scale from poor to excellent, using the categorization scoring method.
- We have analyzed the most promising indicators of theta's influence on student performance and support the use of neural network models to forecast current outcomes through various statistical methods.
- We evaluate the proposed strategy for predicting student performance using actual student data and compare it with different ML models and ML approaches.

Research Question

- Can ML like LR, RF, GB, KNN, and SVM data help us figure out how well training programs work faster?
- Can we use ML to guess how well students will do based on their own data?
- Which is important factors that determinants the performance of students' academic in school and college?

Related Works

Al-Barrak and Al-Razgan [16] examined previous course grades to predict the final grade point average. They used a decision tree algorithm to derive classification criteria from students' transcripts. These rules help identify which classes are essential for a student's final GPA—Iqbal et al. [17] used a ML method to estimate students' GPAs. The dataset was collected from Information Technology University (ITU) in Lahore, Pakistan. They developed a feedback method to determine whether a student understood a specific subject. They proposed a practical approach for applying the Hidden Markov model to predict student performance in particular classes. The ML classifiers were evaluated using three metrics: mean absolute error (MAE), MSE, and RMSE. Zhang et al. [18] adjusted the parameters of the GBDT classifier to forecast students' final grades on graduation theses in Chinese universities. GBDT outperformed KNN, SVM, RF, DT, LDA, and Adaboost in overall prediction accuracy by using AUC metric.

Hilal Almarabeh [19] evaluated several classifiers for assessing students' work. This article compares five ML classifiers. There are several options, including ID3, J48, NN, Naive Bayes, and Bayesian Networks. Weka is used to run all these algorithms in the tests. The data is based on 225 cases, each with 10 attributes, stored in the college database. Jie Xu et al. [20] introduced a new ML approach with two key components. The first feature is a stratified structure for making predictions that considers how students' performance trends change over time. The dataset includes information on 1,169 UCLA undergraduates majoring in Aerospace and Mechanical Engineering. Al-Shehri et al. [21] examined the performance of SVM and KNN, two supervised learning classifiers, using 33 features collected from data at the University of Minho. The initial step in the statistical analysis involved converting the dataset from its nominal to numerical form. The data originated from reports and surveys from two Portuguese educational groups. The dataset consisted of 33 features and 395 instances, with the features distributed as follows: four nominal, 13 binary, and 16 numerical attributes. The experiment utilized the Weka tool to test the algorithms on various datasets. Alowibdi, J [22]. investigated the use of advanced learning analytics to predict student performance in academic settings. They expressed disapproval of Pakistani students who received scholarships. This study evaluated the Bayes Network and Naive Bayes generative models alongside the C4.5, SVM, and CART discriminative models. The predictor's performance was assessed based on its precision, recall, and F-score. Data from 3,000 students collected between 2004 and 2011 was reduced to 776 by removing duplicates and performing pre-processing. Of these, 691 students graduated, while 86 did not. The 33 features were categorized into four groups: family income, family spending, student details, and family assets.

Yağcı [23] proposed an innovative machine learning model for forecasting undergraduate final exam scores based on midterm results, assessing its effectiveness compared to other ML algorithms. Likewise, Niyogisubizo et al. [24] employed an ensemble learning approach to predict university student attrition rates. In a separate study, Huang et al. [25] developed a system that forecasts students' performance in elective courses by analyzing the grade distributions of senior cohorts who previously enrolled in the same courses. Tomasevic et al. [26] conducted a detailed analysis and comparison by systematically improving existing data sources and applying advanced supervised machine learning techniques. Research shows that increasing data volume helps identify patterns in educational data mining. Alshanqiti et al. [27] applied a regression model to examine factors influencing student performance and academic grades. In addition to ML methods, many studies have used deep learning (DL) techniques. Unlike recent studies that predict student performance with machine learning, Akour et al. [28] focused on validating the accuracy of a convolutional neural network-based deep learning algorithm for predicting student grades.

A number of researchers have also studied the topic of predicting dropouts. The authors of [18] designed algorithms to identify students at risk at an American school at the start of the school year using performance data collected during the semester. In the first semester, students' attendance is also evaluated to determine their likelihood of dropping out. Authors of [29] contributed to this area. To enhance the development of retention programs, the authors of [30] use the uplift modeling framework to address the prevention of student dropouts and to quantify the benefits of interventions, such as tutorials. Using data collected at the end of the first year of a student's academic course at the University of Porto in Portugal, well-known data mining classification algorithms predicted students' overall academic success in [31]. The authors of [32] can identify students who are likely to drop out or perform worse than

usual by examining their past grades and combining predictions from different feature groups and models. Using tree-based classification models, the authors of [33] describe two methods for early identification of students at risk of failing or underperforming in an online university statistics course.

Methodology

Figure 2 is a block diagram illustrating the proposed framework for classifying and predicting student performance, outlining the processes used to achieve our results. This

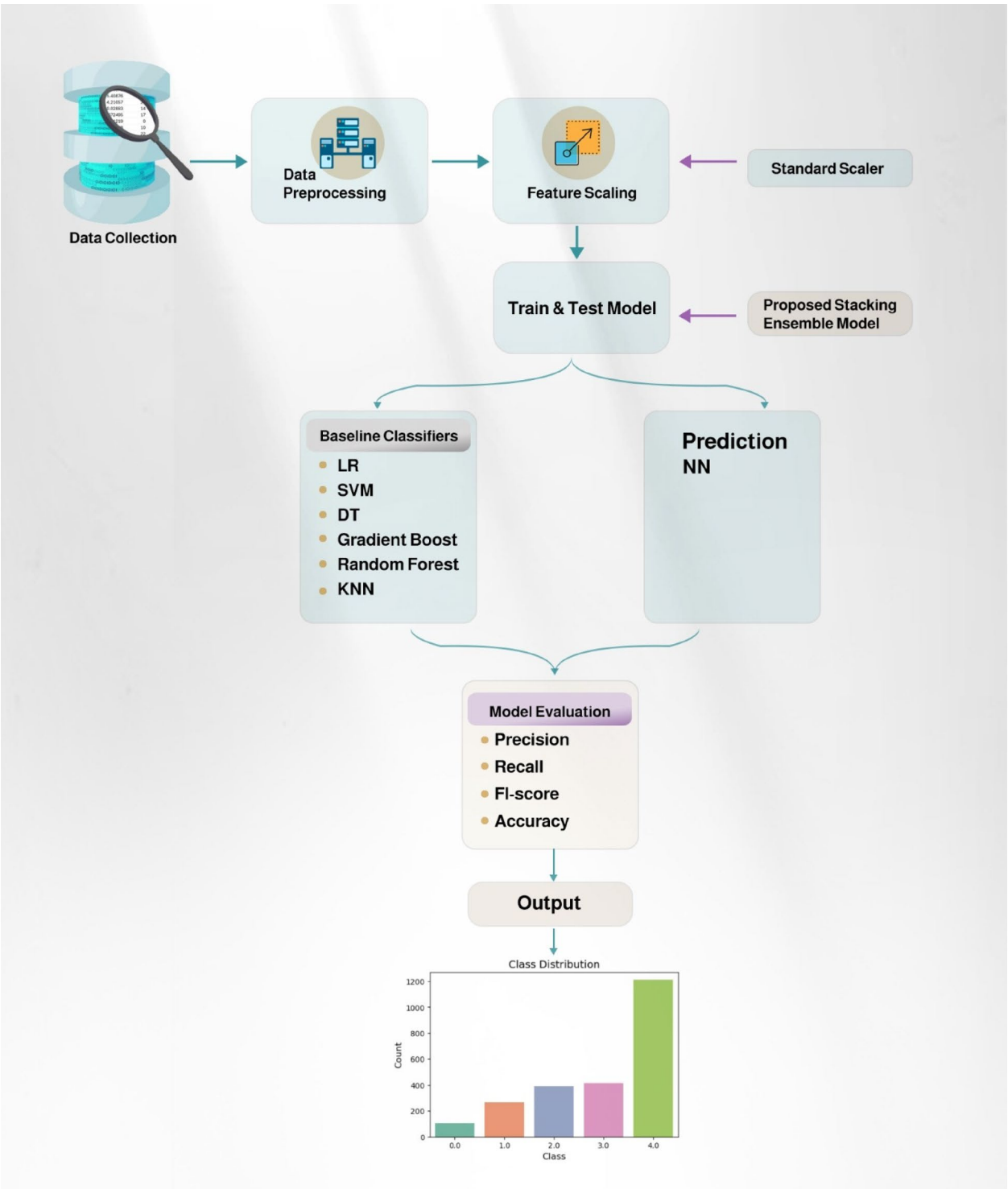


Fig. 2 Framework of the proposed system

Table 1 Features of the dataset

Name of features	Types
#Student_ID	Numerical
#Age	Numerical
#Gender	Categorical
#Ethnicity	Numerical
Parental_Education	Numerical
#Study_Time_Weekly	Numerical
#Absences	Numerical
#Tutoring	Numerical
#Parental_Support	Numerical
#Extracurricular	Numerical
#Sports	Numerical
#Music	Numerical
#Volunteering	Numerical
#GPA	Numerical
#GradeClass	Numerical

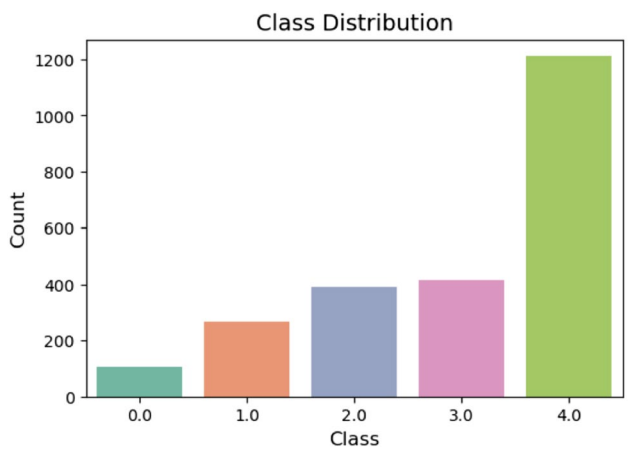


Fig. 3 Classes of datasets

section elaborates on those stages. The following sections detail each stage.

Dataset

This dataset includes information on 2,392 high school students, covering their demographics, study habits, parental involvement, extracurricular activities, and academic achievement. It was sourced from the Kaggle database. The target variable, GradeClass, categorizes students' grades, supporting statistical analysis, predictive modeling, and educational research on a large dataset. The features of the dataset are illustrated in Table 1.

Figure 3 illustrates how grades are divided into groups, from A (0) to F (4). The x-axis displays the different classes, and the y-axis shows the frequency of each class. There is a peak at grade 4 (F), with over 1,200 occurrences, indicating that many students received failing grades.

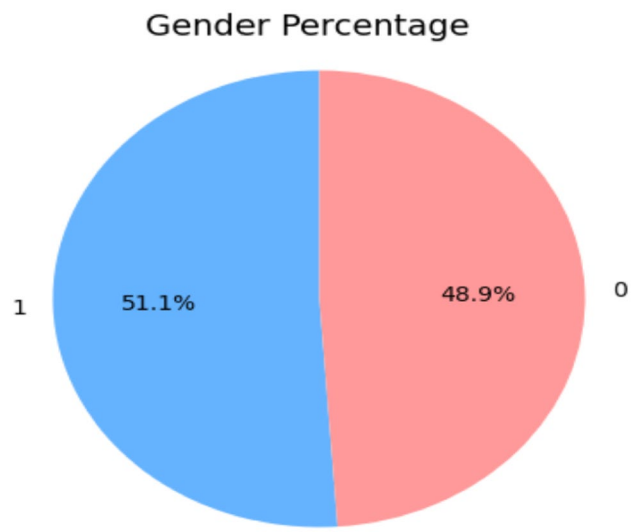


Fig. 4 gender percentage

Preprocessing

Normalization

The Min-max normalization is a prevalent approach for data normalization. All values are transformed into a decimal range between 0 and 1, with the minimum value assigned 0 and the maximum value assigned 1.

$$z_n = \frac{x - x_{min}}{x_{max - x_{min}}} (New_{max_x} - New_{min_x}) + New_{min_x} \quad (1)$$

Features Analysis

The gender distribution pie chart is presented in Fig. 4, showing that the blue and pink categories are nearly evenly split. The blue portion, which accounts for 51.1% of the total, likely represents one gender, while the remaining 48.9% represents the other. This indicates that one gender is significantly more common than the other, revealing a gender imbalance in representation. The approximate percentages can improve analyses of gender representation across different contexts by highlighting an essentially even distribution.

Figure 5 shows the number of people aged 15 to 18. The bars represent the count at each age, with ages 15.5, 16.5, 17.5, and 18.5 having the highest counts, each exceeding 600. The line graph over the bars displays a sinusoidal pattern, indicating that the frequency varies. This pattern suggests that people of certain ages are overrepresented, possibly due to cohort effects or enrollment patterns.

Figure 6 shows the number of people from different ethnicities, ranging from 0 to 3. The x-axis represents the various ethnic group categories, while the y-axis indicates

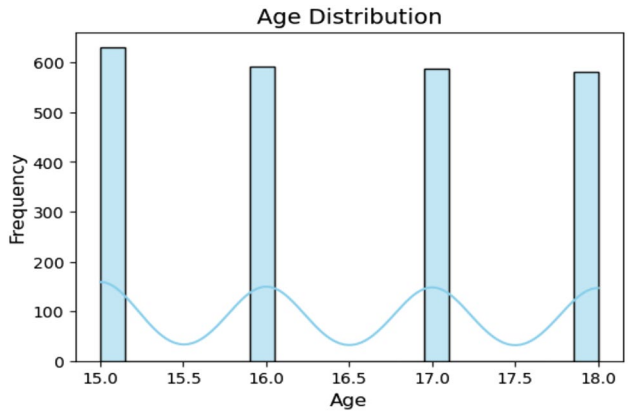


Fig. 5 Age of dataset

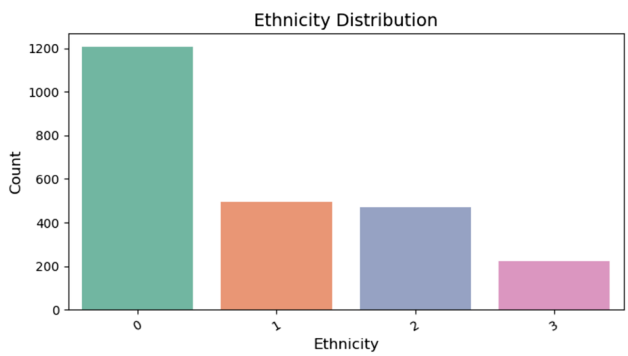


Fig. 6 Ethnicity of the dataset

the number of individuals in each group. Category 0 has over 1200 people, as shown by the tallest bar, making it the largest ethnic group in this dataset. The bar for category 3 is much narrower, indicating the fewest people. Categories 1 and 2 have moderate numbers, with category 1 slightly higher than category 2.

Figure 7 shows the various factors influencing a specific result. It illustrates the importance of each feature as identified by a RF model. The y-axis lists the attributes, while the x-axis represents the significance score. With an importance value of 0.4, GPA clearly stands out as the most important factor, highlighting its role in making predictions. Absenteeism is less important than GPA but still plays a role. Age, ethnicity, gender, sports, tutoring, extracurricular activities, volunteering, and music have minimal impact, whereas parental education, weekly study time, and parental support are more influential. Although other characteristics matter to different extents, this distribution indicates that academic success (GPA) significantly affects the expected outcome.

$$Importna\ (f_t) = \sum {}^{T}_{t=1} \sum {}_{n\in\ t} \Delta\ I_n\ (f_i) \tag{2}$$

Features with higher importance values contribute more significantly to model prediction accuracy. The decrease in Gini impurity at node n when the feature $f_y$is employed for splitting is represented $\Delta\ I_n\ (f_i)$ by prediction accuracy is greatly improved by features with greater significance ratings.

Classification Algorithms

Random Forest

A RF method is ML approach that uses numerous decision trees to integrate their predictions and create a more accurate and dependable output. As a supervised learning approach,

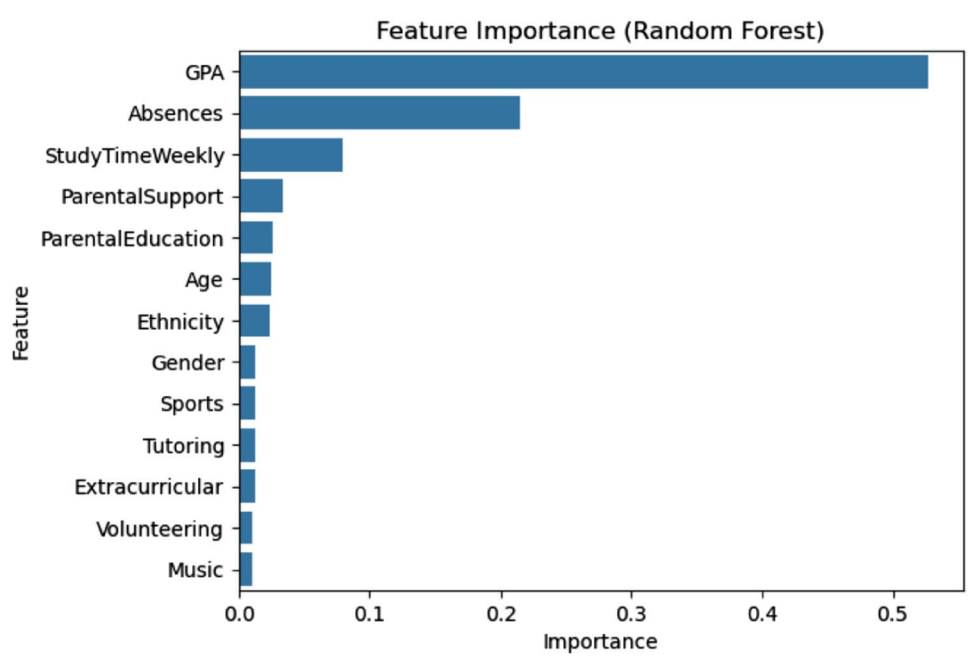


Fig. 7 Important features of the dataset

it can be applied to both regression and classification tasks. The RF Regression method is used to estimate numerical values in regression problems. It predicts continuous outputs by averaging the results from many decision trees. In RF Regression, several decision trees are created, each trained on a different random subset of the data. The first step is Bootstrap sampling, which involves selecting random rows of data with replacement to create separate training datasets for each tree. Model diversity is maintained through feature sampling, where each tree is built using only a randomly selected subset of features. The structure of RF is shown in Fig. 8.

Gradient Boosting

The GB approach is an ensemble learning method suitable for both regression and classification tasks. This boosting technique combines multiple weak learners to create a stronger prediction model. It aims to correct the errors made by earlier models by training them sequentially [34–36]. Gradient boosting adds a new model to the existing ones by using gradient descent to reduce the loss function of the previous model. The loss function could be mean squared error or cross-entropy. The algorithm identifies the relationship between the predictions and the gradient of the loss function at each cycle. Then, it trains a new weak model to minimize this gradient as much as possible. All the models' predictions are combined to form the ensemble, and the projections of the latest model are added to it. This process continues iteratively until a specified condition is satisfied. Figure 9 displays the structure of GB method.

Evaluation of Metrics

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \times 100 \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \times 100 \tag{5}$$

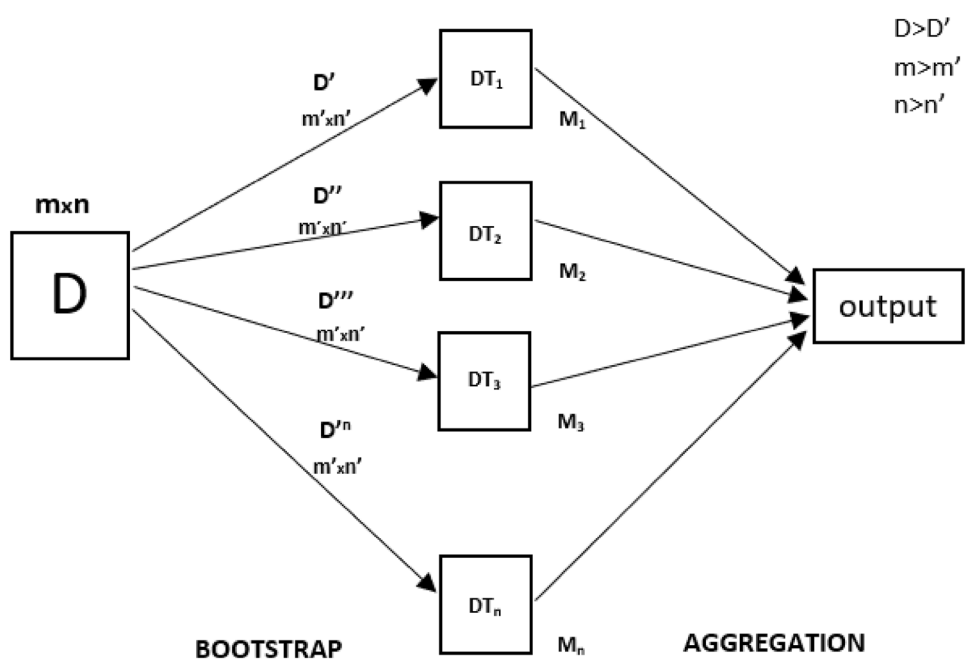
$$F1 - sore = 2 \times \frac{Precision \times Recall}{Precision + Recall} \times 100 \tag{6}$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_{i,exp} - y_{i,\,pred})^2 \tag{7}$$

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(y_{i,exp} - y_{i,pred})^2}{n}} \tag{8}$$

$$R\% = \frac{n\left(\sum_{i=1}^{n} y_{i,exp} \times y_{i,\,pred}\right) - \left(\sum_{i=1}^{n} y_{i,exp}\right)\left(\sum_{i=1}^{n} y_{i,\,pred}\right)}{\sqrt{\left[n\left(\sum_{i=1}^{n} y_{i,exp}\right)^2 - \left(\sum_{i=1}^{n} y_{i,exp}\right)^2\right]\left[n\left(\sum_{i=1}^{n} y_{i,pred}\right)^2 - \left(\sum_{i=1}^{n} y_{i,pred}\right)^2\right]}} \times 100 \tag{9}$$

figure. 8 Generic structure of RF



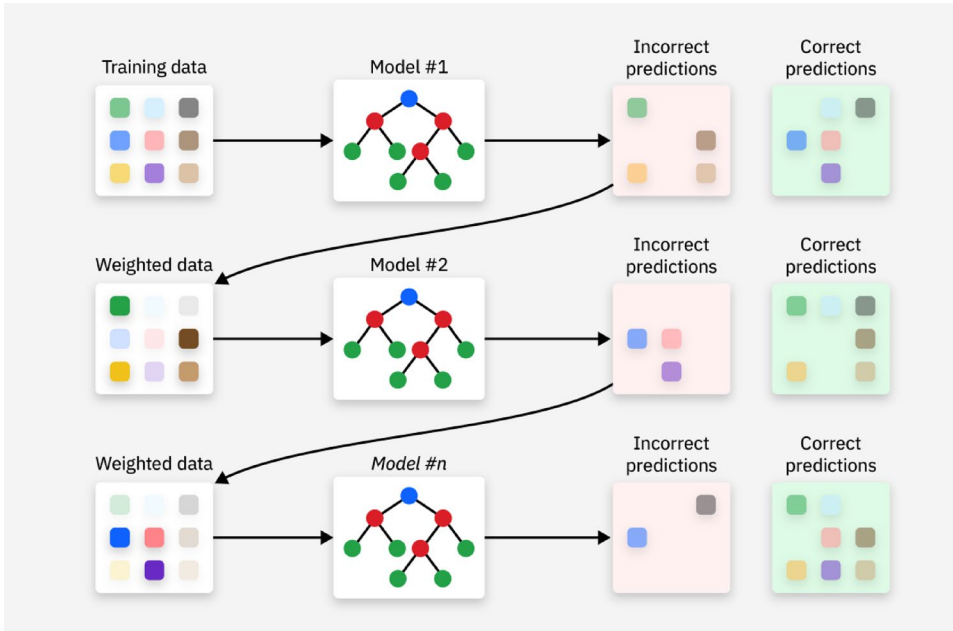


Fig. 9 Generic structure of GB

Table 2 Results of LR method

Class Name	Precision (%)	Recall (%)	F1 Score (%)	Support
0.0	40	19	26	21
1.0	63	54	58	54
2.0	64	78	70	78
3.0	79	65	72	83
4.0	91	79	94	243
Accuracy		80		
Macro Avg	67	63	64	243

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_{i,\,exp} - y_{i,\,pred}\right)^2}{\sum_{i=1}^{n} \left(y_{i,\,exp} - y_{avg,\,exp}\right)^2} \tag{10}$$

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{k=1}^{n}(y_{i,\text{exp}} - y_{i,pred})^2}}{y_{i,pred}} \tag{11}$$

Experimental Analysis

In this experiment. We utilized the standard dataset obtained from Kaggle, which contains the most pertinent indicators of student performance. In this segment, we employed Python programming for the classification and prediction of student performance. The dataset was divided into 80% for training and 20% for testing. This implementation utilized many environments, including software and hardware, such as diverse libraries for ML and PC_i5, along with 16 GB of RAM, to achieve high accuracy. Different ML models were employed to predict student performance.

Classification results

Results of LR

The LR model's results, shown in Table 2, reveal that prediction accuracy varies significantly with student performance (0.0 to 4.0). Class 0.0 is particularly challenging for the model, as it recalls only 19% of the students, with an F1 score of 26% and an accuracy of 40%. These figures suggest that the model struggles to identify students with the lowest performance reliably. Class 2.0 produces better results (64% accuracy, 78% recall, 70% F1), while Class 1.0 is less accurate in predictions (63% precision, 54% recall, 58% F1). As the number of classes increases, performance consistently improves. Class 4.0 has the highest predictive accuracy, with 91% precision, 79% recall, and a 94% F1 score. Conversely, Class 3.0 has 79% precision and a 72% F1 score. The model performs well for students who excel academically. The macro average results (67% precision, 63% recall, 64% F1) indicate that the model performs better at identifying medium- to high-achieving students than low-achieving ones, despite an overall accuracy of 80%.

The result of DT methods' classification performance for students is shown in Fig. 3, with score markings (class names 0.0 to 4.0). The DT model has a solid overall accuracy of 86%. This is supported by a balanced weighted average of 78% across precision, recall, and F1 score. It's difficult to identify students who are failing because the model doesn't perform very well for those with the lowest scores (class 0.0) (56% accuracy, 43% recall, 49% F1). Class 2.0 produces excellent results (84% accuracy, 94% recall, and 88% F1), indicating that mid-range performances are classified

Table 3 Results of DT method

Class Name	Precision (%)	Recall (%)	F1 Score (%)	Support
0.0	56	43	49	21
1.0	72	85	78	54
2.0	84	94	88	78
3.0	88	80	84	83
4.0	92	89	90	243
Accuracy	86			
Weighted	78	78	78	243

Table 4 Results of the RF method

Class Name	Precision (%)	Recall (%)	F1 Score (%)	Support
0.0	100	33	50	21
1.0	83	83	83	54
2.0	88	95	91	78
3.0	89	89	89	83
4.0	94	97	96	243
Accuracy	91			
Weighted	91	91	90	243

Table 5 Results of gradient boosting method

Class Name	Precision (%)	Recall (%)	F1 Score (%)	Support
0.0	91	48	62	21
1.0	91	89	90	54
2.0	88	94	92	78
3.0	88	89	89	83
4.0	94	97	95	243
Accuracy	92			
Weighted	92	92	92	243

with high confidence. Class 1.0 has only 72% accuracy and 78% F1. Class 3.0 continues to perform well, achieving 88% accuracy, 80% recall, and 84% F1 score. With an accuracy of 92%, a recall of 89%, and an F1 score of 90%, Class 4.0, which includes the top students, outperforms all other classes. Based on this data, the DT model effectively identifies students with average to above-average grades (Table 3).

The RF model results are shown in Table 4; the results of RF indicate it is very effective at predicting student scores. This is demonstrated by balanced weighted averages of 91% for precision, 91% for recall, and 90% for the F1 score. It is difficult to identify students who are not doing well when the model achieves 100% accuracy but only 33% recall (50% F1) for the students who are struggling the most (class 0.0). The model performs well in predicting the students it has identified, but it doesn't catch most of them. However, performance remains reliable for other groups. Class 2.0 has an accuracy of 88%, a recall of 95%, and an F1 score of 91%, indicating it is effective at identifying intermediate performers. Class 1.0 consistently scores 83% across all metrics. Class 3.0 shows good results with 89% recall, 97% precision, and a 96% F1 score, while the top performers in class 4.0 have nearly perfect metrics (94% precision, 97% recall, and 96% F1). Research indicates that RF can reliably identify students who are performing well academically, but other approaches, such as balancing methods or feature tuning, may be necessary to identify students with poor grades more quickly.

Table 6 Results of KNN method

Class Name	Precision (%)	Recall (%)	F1 Score (%)	Support
0.0	23	14	18	21
1.0	40	30	34	54
2.0	42	56	48	78
3.0	46	35	40	83
4.0	84	89	86	243
Accuracy	64			
Weighted	63	64	63	243

Table 7 Results of SVM method

Class Name	Precision (%)	Recall (%)	F1 Score (%)	Support
0.0	80	19	31	21
1.0	66	65	65	54
2.0	65	72	68	78
3.0	67	71	69	83
4.0	92	94	93	243
Accuracy		80		
Weighted	80	80	79	243

The overall accuracy of the GB model in predicting student scores is 92%, as shown in Table 5. The weighted averages for precision, recall, and F1 score are all 92%. The model performs well with the lowest-performing group (class 0.0), achieving a recall of 48% (F1 score of 62%) and a precision of 91%. This indicates it correctly identifies students who are failing when expected, but it misses many of them. For class 1.0, the best balance is observed with 91% precision, 89% recall, and 90% F1 score. Class 2.0 shows the highest accuracy, with 88% precision, 94% recall, and 92% F1. Class 3.0 demonstrates the most incredible consistency, with 88% precision, 89% recall, and 89% F1. Performance remains strong in the other courses, with the top-performing students (class 4.0) predicted with exceptional accuracy—94% precision, 97% recall, and a 95% F1 score. GB is a valuable tool for predicting student performance, as it effectively distinguishes between students with medium to high scores.

Table 6 shows that the KNN results, with an overall weighted accuracy, recall, and F1 score of approximately 63–64%, are insufficient for predicting student scores compared to other models. The model struggles significantly with students who do not perform well. Class 0.0 achieves 23% accuracy, 14% recall, and an 18% F1 score, while class 1.0 performs slightly better with 40% accuracy, 30% recall, and 34% F1 score. Class 2.0 shows better average performance, with 42% accuracy, 56% recall, and 48% F1, compared to class 3.0, which highlights the difficulty in

distinguishing between regular students, with 46% precision, 35% recall, and 40% F1. The model performs particularly well with high achievers (class 4.0), demonstrating a bias toward accurately identifying top students, as reflected by an accuracy of 84%, recall of 89%, and an F1 score of 86%.

The SVM method is highly effective in predicting student scores, as shown in Table 7, achieving an average accuracy of 80%, a recall of 80%, and an F1 score of 79%. However, its effectiveness varies across academic disciplines. The model's ability to identify students is very poor for the lowest-performing students (class 0.0). The model is 80% accurate, but it only recognizes 19% of the students (31% F1), meaning it finds some students but misses most. Both Class 2.0 and Class 1.0 show improvements in performance, with similar values (65–68% F1), indicating they are reliable enough for students performing well to average. The results for Class 3.0 are consistent, with 67% F1, 71% recall, and 71% accuracy. Class 4.0 yields better results because the model excels at identifying high-achieving students (92% accuracy, 94% recall, and 93% F1). From an academic perspective, SVM performs well for high achievers and overall accuracy, but it struggles to identify students who are not doing well.

Figure 10 displays the confusion matrices for the proposed ML model. The LR model performs best in class 4, correctly identifying 236 instances, demonstrating its strong ability to predict this category. Class 2 has the most errors, with 15 cases misclassified as class 1 and 7 as class 0, indicating difficulty in distinguishing these classes. This issue is particularly evident for classes 1, 2, and 3. Misclassification is notable in Class 3, where 17 cases are incorrectly assigned to Class 2. The Logistic Regression model effectively detects many cases in class 4. The classification results for Class 4 highlight the performance of the DT model, with 219 correct predictions across five groups. Class 2 shows strong categorization with 69 true positives. Class 1 has 8 true positives, along with some misclassified as class 2, revealing challenges in differentiating between classes 1 and 3, as reflected by the higher misclassification rates. Additionally, Class 3 experiences several errors, with four instances wrongly labeled as class 2 and twelve as class 1.

The Random Forest model demonstrates its ability to categorize data using five different types of data. It accurately

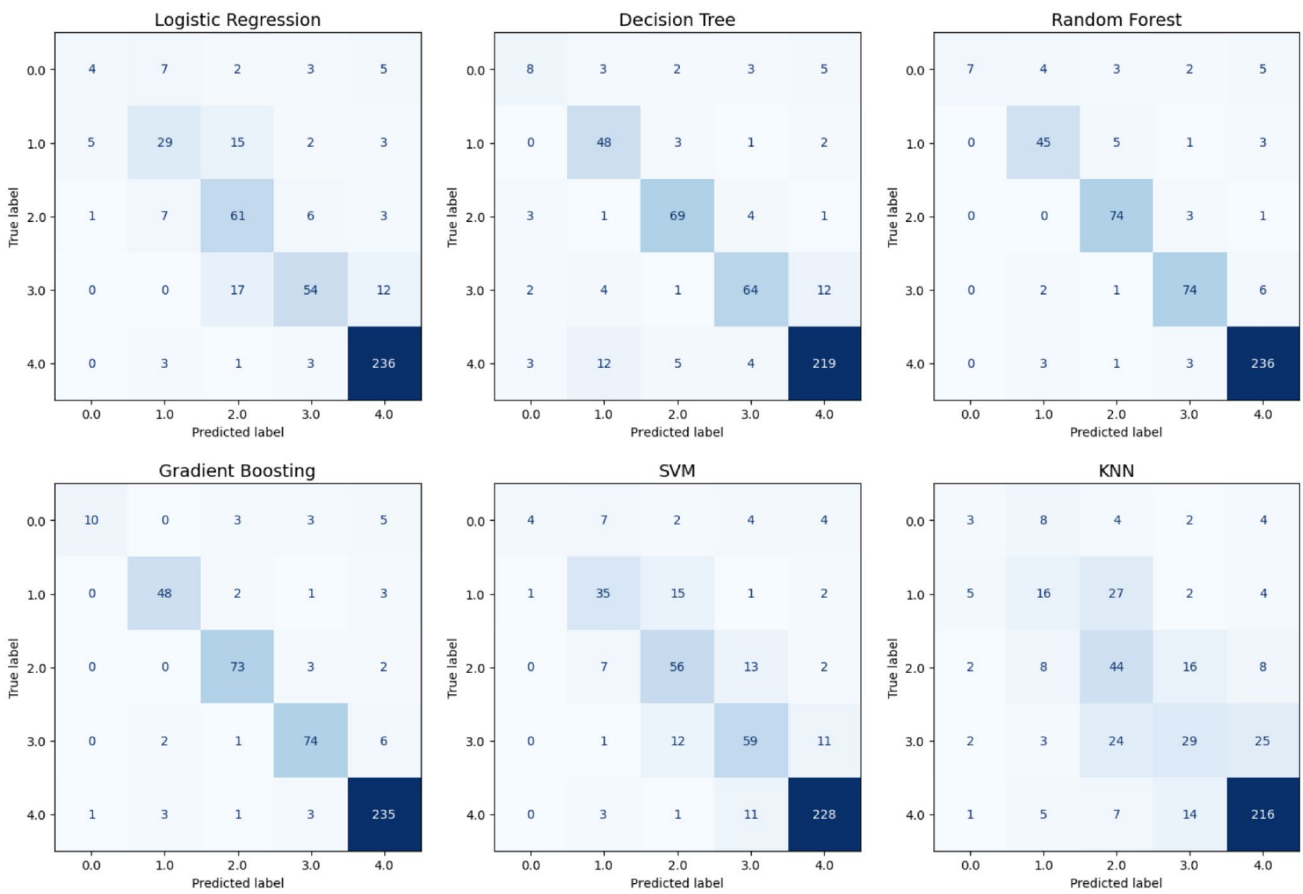


Fig. 10 Confusion matrix of ML models

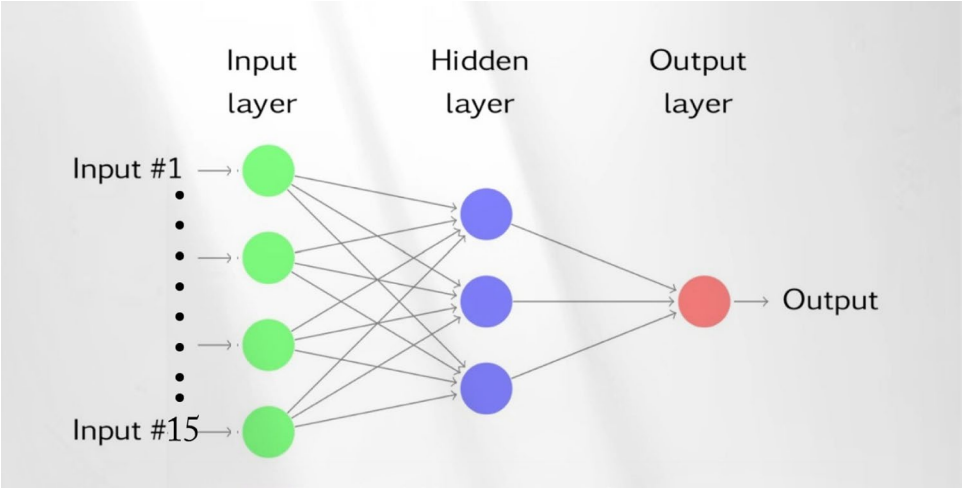


Fig. 11 Structure of NN

Table 8 Prediction results using NN model

Metrics	Values
MSE	0.0665
RMSE	0.2579
R^2 Score	93.27
MAPE	0.7176
Mean of predictions	0.0358
Standard deviation	0.9506

predicts class 4 by correctly identifying 236 cases. Class 2 also performs well, with the model correctly recognizing this group in 74 cases. However, it struggles with classes 1 and 3. Class 1 has been misclassified as class 2 five times, and as class 0 four times. The model faces difficulty accurately classifying Class 3, managing only 1 true positive from multiple predictions. Overall, the Random Forest model shows high accuracy in some classes, especially classes 2 and 4, but makes errors in others. Class 4 exhibits exceptional performance, indicated by 235 true positives, highlighting its effectiveness in categorizing this dataset. Class 2 also performs well with 74 true positives. Conversely, the model has trouble with class 1, misclassifying 3 cases as class 2 and 10 as class 0. The SVM model achieves 238 true positives in class 4 and 35 in class 2, which are strong scores. However, class 1 exhibits misclassifications, with five instances predicted as class 0 and thirteen as class 2, indicating challenges in distinguishing between these categories. KNN performs commendably with 216 accurate predictions in class 4. However, it faces issues with class 1, where eight students were misclassified as class 0 and forty-four as class 2. This highlights KNN's reliance on local data points for categorization and its sensitivity to noise.

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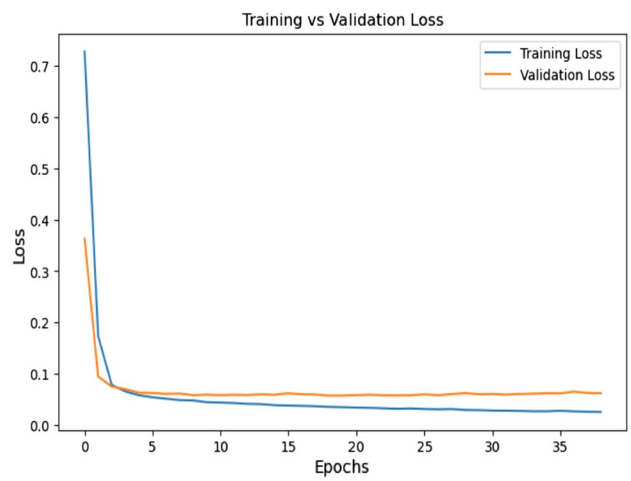


Fig. 12 Prediction loss of the NN model

Prediction Results

The neural network (NN) is structured hierarchically. The layer with predictors is at the bottom, while the layers with predictions are at the top. Hidden neurons in intermediate levels are also possible. Figure 11 shows the NN implementation for predicting students' performance at this school. The weights refer to the coefficients linked to these predictors. The inputs are linearly combined to produce the projections. A learning algorithm that minimizes a cost function, such as the MSE, is used to determine the weights within the neural network. This model works well for regression tasks because it has a sequential network design with an input layer of 64 neurons activated by ReLU and a single neuron in the output layer with a linear activation function. The program is compiled using the Adam optimizer and the mean squared error (MSE) loss function. Parameters monitor validation loss and set levels for patience and verbosity. Early stopping prevents overfitting. In this section, we use NN to predict students' performance, as shown in Table 8.

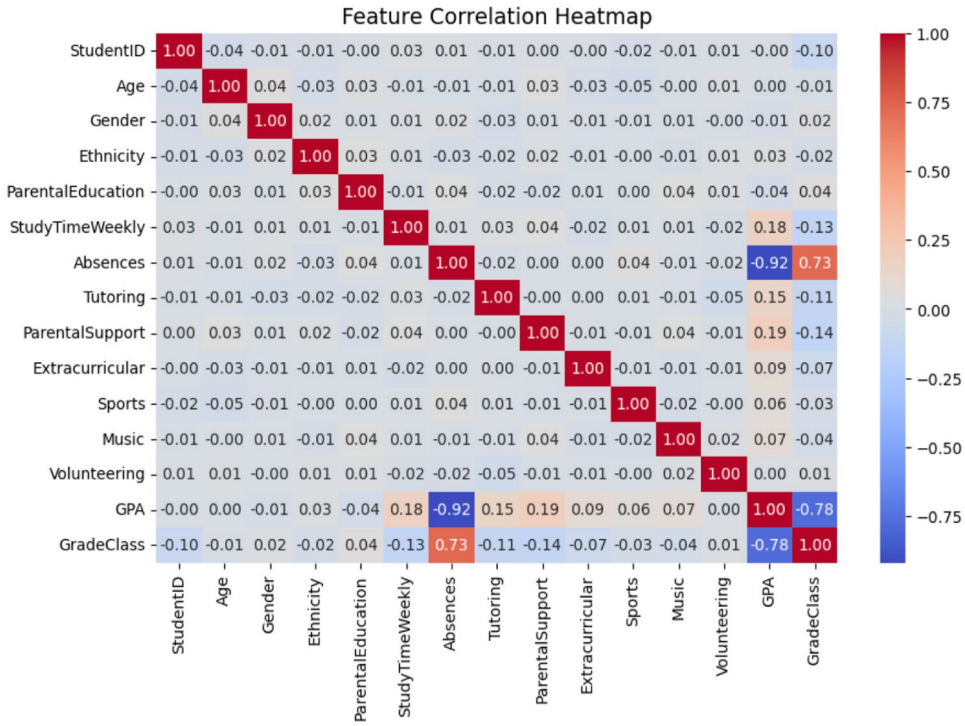


Fig. 13 correction features of the dataset

The final data showed that the neural network had completed training, with a loss of 0.0665 and a mean absolute error (MAE) of 0.2008, as shown in Fig. 12. The model fits the training data well, indicated by the low loss value, which suggests it understands the underlying patterns quite well. The MAE of 0.2008, representing a slight average absolute difference between actual and predicted values, further confirms strong predictive accuracy. Overall, these signs indicate that the training was successful, which bodes well for the model's performance on new data.

The feature correlation heatmap shows different properties of the dataset in Fig. 13. The traits include StudentID, Age, Gender, Ethnicity, Parental Education, Weekly Study Time, Absences, Tutoring, Parental Support, Extracurricular Activities, Sports, Music, Volunteering, GPA, and Grade Class. There is a strong positive correlation (0.73) between Grade Point Average and class rank, indicating that students with higher GPAs tend to have higher ranks. Additionally, there is a strong link (0.92) between GPA and extracurricular activities, suggesting that participating in these activities can significantly boost academic performance. Volunteering has a moderate correlation (0.54) with GPA, while Weekly Study Time shows a mild positive correlation (0.62) with GPA, further emphasizing the connection between longer study hours and better academic results. Age, gender, and ethnicity are not strongly associated with GPA, meaning they have less impact on academic success compared to factors like tutoring and extracurricular activities. The heatmap highlights all the activities students engage in outside of class and how their study habits influence their grades.

Table 9 Final accuracy of ML models

Models	Accuracy
Logistic regression	80.16
Random forest	91.02
Gradient boosting	91.85
KNN	64.30
SVM	79.74

Discussion

It's impossible to overstate the importance of education today. Education helps us evaluate various claims made by different groups and determine which ones we can trust. It also helps individuals become moral members of society. A significant amount of research has studied how AI approaches can be applied in schools to improve academic performance. This study aims to identify the factors that affect students' performance and to identify those students who may be at risk. This study employed Drive-ML methodologies to forecast and categorize student performance in educational settings.

Table 9 presents the final performance of the ML models in classifying students' scores for evaluating their academic achievement. The final model accuracies reveal significant differences in how well they predict student grades. The GB model was the most accurate ensemble-based tree model,

with an accuracy of 91.85%. Next was the RF model, with an accuracy of 91.02%. This indicates that these models best capture the complex patterns in the dataset. Ensemble methods offer greater flexibility; however, a simple linear model can yield impressive results, as demonstrated by the LR model's notable performance of 80.16%. SVM achieved an accuracy of 79.74%, comparable to that of boosting methods. This shows that it is consistently effective, though not highly effective, at handling nonlinear decision boundaries. Conversely, KNN's ineffectiveness in this context was evident, with only 64.30% accuracy. This is likely due to its sensitivity to high-dimensional scaling and difficulty in generalizing across different groups of students. These findings suggest that RF and GB are the best models for predicting academic success. Logistic Regression and Support Vector Machines serve as reliable baseline techniques. However, KNN does not perform as well as it appears and is likely not the best method for predicting academic outcomes.

Figure 14 shows ROC curves of ML models, showing how well they can classify data into different groups, such as DT, LR, RF, GB, SVM, and KNN. An area under the curve (AUC) of 0.75 for logistic regression indicates a substantial difficulty in differentiating between groups. The DT has an AUC of 0.80, which means that the curve changes, which could mean that it works well for some classes, but also means that it might be overfitting. The RF model has an AUC of 0.92, which shows that it has very high true positive rates and is very good at telling the difference between classes, especially for class 4 GB, with an AUC of 0.88, showing strong overall performance, especially when it comes to describing the difference between good and bad situations. KNN has an AUC of 0.76, indicating that it may not always perform optimally and is susceptible to noise and misclassification. On the other hand, SVM achieves an AUC of 0.90, which means it can effectively distinguish between classes. SVM and RF appear to be the best models overall, but LR and KNN were achieving less.

The performance comparison shown in Fig. 15 includes the F1 scores, recall, precision, and accuracy of LR, DT, RF, GB, SVM, and KNN. SVM performs well, especially in terms of accuracy and F1 score, but Random Forest consistently produces better results overall, indicating it is effective at classifying applications. Gradient Boosting outperforms other methods, particularly regarding accuracy, and manages positive events well. The Decision Tree performs adequately overall but struggles with remembering details, which makes it challenging to identify all the optimal situations. Logistic Regression and KNN did not

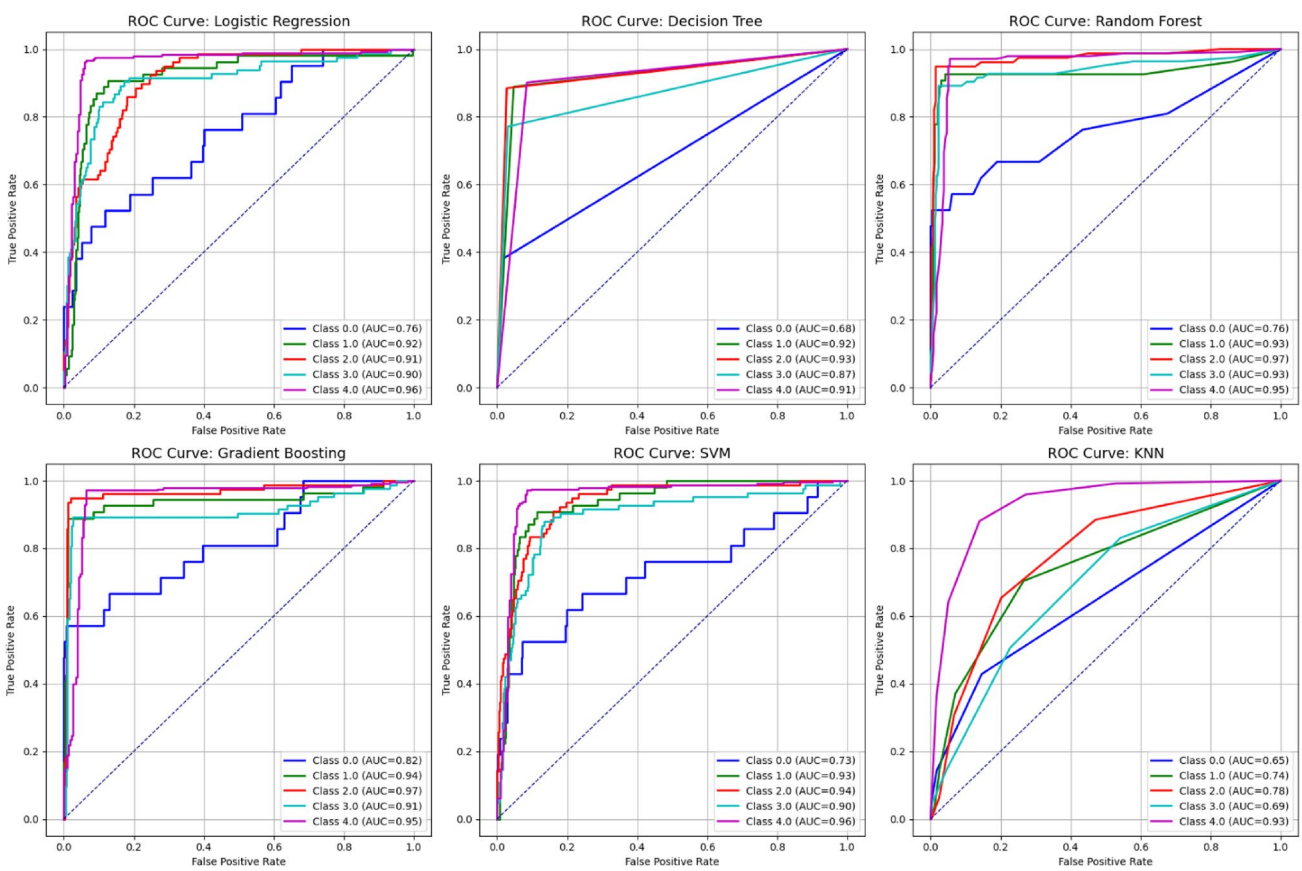
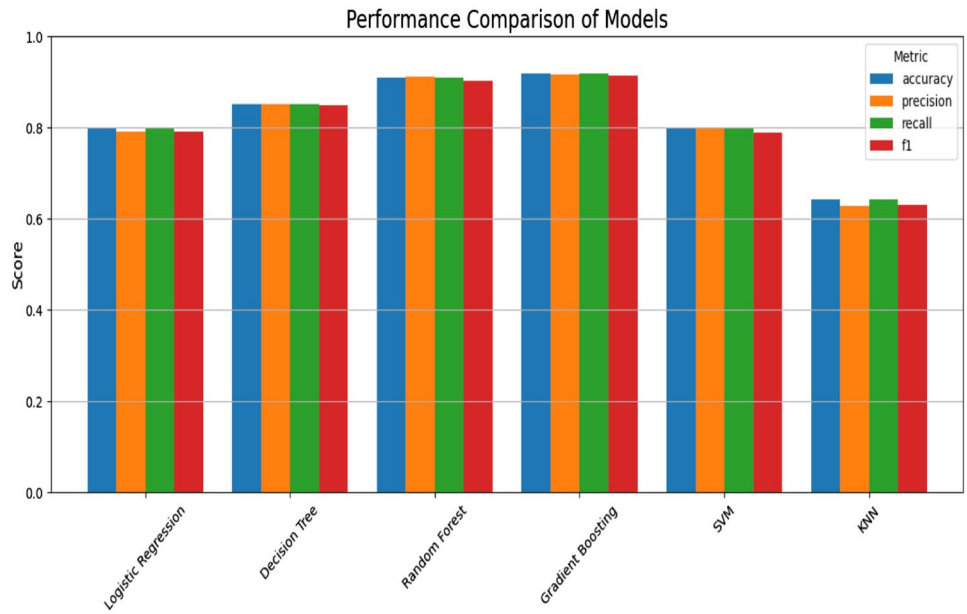


Fig. 14 ROC of ML models

Fig. 15 performance of ML models



perform well, particularly in terms of accuracy and F1 score, and therefore require improvement.

Conclusion

This study investigates both individual and normative characteristics of students to forecast their academic performance through an analysis of these traits. The conclusion is as follows:

> The main goal of this study is to determine how well students will perform in school so they can improve their academic success.

- This project proposes an AI-based system to assist teachers in grading students' work. In this study, we used the standardized dataset from 2,392 high schools.
- In this study, we used multiple ML models, such as Random Forest (RF), Gradient Boosting (GB), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), to classify student scores for performance evaluation.
- We used a neural network model and various statistical tests to predict students' performance.
- Ultimately, the system observed that GB and RF achieved a very high accuracy rate of 91–92%. I noticed that this model was effective for analyzing and classifying performance with this dataset.

We use real student data to evaluate the proposed model of academic success. The results show that our method effectively predicts student performance. By tracking their students' progress and adjusting teaching strategies based on what they learn, teachers can better meet their students' needs. To efficiently allocate resources to student groups with different performance levels, educational administrators need a clear understanding of the current distribution of student performance and the trends in their future performance. In future scope, using social and psychological, and environmental factors for making predictions more accurate.

Authors' Contributions Asma Abdulmana Alhamadi has read and agreed to the published version of the manuscript.

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Data Availability https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset.

Declarations

Conflict of interest The author declares no conflict of interest.

Research Involving Human and Animals Participants This type of study is non-human and animal subjects research.

Consent for Publication The Author has read and agreed to the publication of the manuscript.

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